(a) Theory–experiment pairs → shared spectral space

64 queries · 96 references · τ = 0.1

Theory y
3501 points
500–4000 cm⁻¹

Proxy (c)

Theory view ỹ

Experiment x

Spectrum Transformer
16-point patches / stride 8
436 tokens → 3 layers × 4 heads
Coverage-weighted pool → h¹²⁸

Shared weights across both domains

Projection
+ L2 norm
128 → 64

u

u

64-D unit vectors

Shared latent space

● queries ■ theory refs

Pairing and view metadata

Canonical identity → positive-pair mask

Source label
QME14S / fg26

Source → proxy views (c)
Identity → matches (b, d)
Coverage → encoder (e)

Theory refs
96 spectra

64 matched
+ 32 extra (d)

Spectrum Transformer
Same weights · mask-aware pool

Projection
+ L2 norm

v

(b) Multi-positive domain alignment

96 references

U

V

64 queries

U → V / V → U

Same molecule = positive

Full identity; repeated positives allowed

Sum positive probability mass

Different identities are negatives

Reverse: supported references only

LQ→R = −mean log [Σ positives exp(score) / Σ all exp(score)]

score(i,j) = u(i)·v(j) / τ; positives share full molecular identity.

(c) Source-dependent proxy augmentation

Theoretical curve → **Fixed source proxy** → **Randomized views**

Frequency scaling a, shift b and broadening σ, fitted per source

Train-only pairs calibrate the proxy; base augmentation stays active.

Synthetic queries enter the same contrastive objective in (b).

(d) Training-only hard-negative selection

Real query → Embedding → Training ranks → **Remove same-identity refs** → **Sample one** Top-16 errors

32 extra references per batch; control arms sample randomly.

Epoch 1 uses random extras; hard candidates refresh each epoch.

The extra-reference budget is identical across training arms.

(e) Coverage-aware encoding

Spectrum + mask

Gray = unmeasured

Valid-bin norm

Patch masking

Patch coverage weights

Weighted pool → h¹²⁸

Observed zeros remain valid; missing regions have a separate mask.

Fully missing patches are excluded as attention keys and values.

Partial patches contribute in proportion to their observed coverage.

Ltotal = ½ (LQ→R + LR→Q) · **Proxy views and hard negatives feed this single objective**

INFERENCE Query + mask → shared encoder → u⁶⁴ → Cosine → 10,000 theory spectrum embeddings → **Top 10 → LLM → one candidate index**

Training / validation / test experimental groups are disjoint; proxy fit and mining use training data only.

All curves and matrices are schematic.